

Sho Sakaue

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Education

University of Utah – Bachelor of Science in Computer Science & Applied Math (December 2027)

- **GPA:** 3.97/4.0 Dean's List (all semesters)
- **Mathematics:** Linear Algebra, Numerical Analysis, Probability theory & Statistics, Real Analysis
- **CS:** Machine Learning & Deep Learning, Data Structures & Algorithms, Model of Computation

Skills

- **Programming:** Python, Java, C++, C#, SQL; Object-Oriented-Programming, Multithreading
- **Machine Learning:** PyTorch, Hugging Face, NumPy, scikit-learn, Pandas, PEFT(LoRA & QLoRA)
- **Developer tools:** Git/GitHub, Docker, Linux, HPC/SLURM computing

Work and Research Experiences

Research Assistant — STARS Lab, University of Utah — UROP funded (August 2025 - Present)

- Developed an automated atomistic simulation workflow using ASE to generate defect, surface, grain-boundary, and fracture-related structures for DFT and MLIP training datasets.
- Constructed and curated DFT reference datasets for complex alloy systems, including vacancy, stacking-fault, surface, and grain-boundary configurations to improve MLIP transferability.
- Fine-tuned Machine Learning Interatomic Potentials (MLIPs) using DFT-calculated energies and forces, improving accuracy for defect energetics and mechanical response predictions.
- Validated fine-tuned MLIP predictions against DFT, experimental data, and other MLIP models using structural and thermal properties, including lattice constants, thermal expansion, and stacking fault energies.
- Accelerated structure relaxation from DFT-scale hours to 3–5 minutes per configuration using MLIP-based simulations

Projects

Parameter-Efficient Fine-Tuning of LLM (LoRA/QLoRA) (March 2026 - April 2026)

- Implemented LoRA and QLoRA from the original papers in PyTorch, manually building the low-rank adaptor and quantization pipeline rather than relying solely on high-level libraries.
- Achieved 99% parameter reduction compared to full fine-tuning while maintaining competitive model performance.
- Fine-tuned a llama-2-7B model and benchmarked parameter-efficient methods against base and fully fine-tuned models on single GPU, evaluating accuracy-performance tradeoffs.
- Conducted ablation experiments varying LoRA rank and quantization bit-width, analyzing their impact on accuracy, memory efficiency, and training behavior.

Multiplayer Networked Game & HTTP Server — Software Practice (November 2025 - December 2025)

- Developed a distributed, multiplayer game client using C# and Blazor following the MVC architectural pattern.
- Made a multi-threaded TCP network controller to handle real-time JSON game state updates.
- Managed thread safety and concurrency using locks to prevent race conditions during asynchronous data processing.
- Integrated a MySQL database to store game history and player high scores.

Competition / Hackathon

MIT iQuHACK — Quantum Computer Hackathon (January 31, 2026 - February 1, 2026)

- Analyzed large quantum circuits (50–60 qubits) from QASM files and implemented quantum circuit simulations using Qiskit to identify hidden peak bitstrings and achieved 24th place in the BlueQubit challenge.
- Optimized performance through gate reordering and cancellation to lower circuit complexity and simulation runtime for 10,000 shots by about 30%.
- Designed statistical post-processing techniques to extract peak signal from near-uniform output distributions, including bitstring filtering by Hamming weight and bit-position frequency patterns, and achieved accurate inference with just 10,000 shots instead of exponential brute-force search
- Developed candidate-selection heuristics by comparing Hamming distances across sampled outputs to rank likely peak solutions.